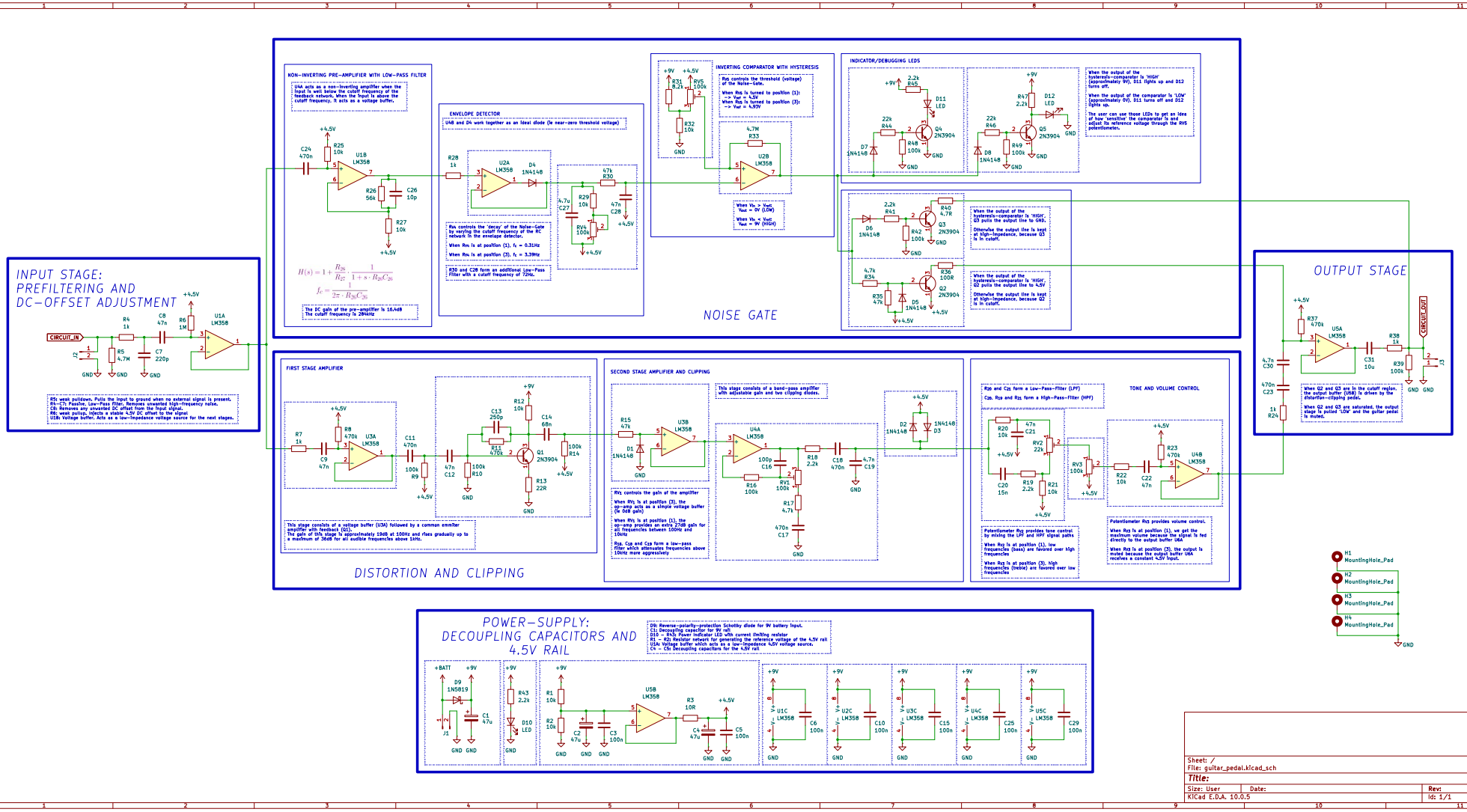

INPUT STAGE:
PREFILTERING AND
DC-OFFSET ADJUSTMENT
R4 1k
C8 47n
R6 1M
U1A LM358
CIRCUIT_IN
R5 4.7M
C7 220p
GND
+4.5V
R5: weak pulldown. Pulls the input to ground when no external signal is present.
R4–C7: Passive, Low-Pass filter. Removes unwanted high-frequency noise.
C8: Removes any unwanted DC offset from the input signal.
R6: weak pullup. injects a stable 4.5V DC offset to the signal.
U1A: Voltage buffer. Acts as a low-impedance voltage source for the next stages.
NON-INVERTING PRE-AMPLIFIER WITH LOW-PASS FILTER
U1A acts as a non-inverting amplifier when the input is well below the cutoff frequency of the feedback network. When the input is above the cutoff frequency, it acts as a voltage buffer.
C24 470n
R25 10k
U1B LM358
R26 56k
C26 10p
R27 10k
$H(s) = 1 + \frac{R_{26}}{R_{27}} \cdot \frac{1}{1 + s \cdot R_{26} C_{26}}$
$f_c = \frac{1}{2\pi \cdot R_{26} C_{26}}$
The DC gain of the pre-amplifier is 10.4dB
The cutoff frequency is 2844Hz
ENVELOPE DETECTOR
U4B and D4 work together as an ideal diode (ie near-zero threshold voltage)
R28 1k
U2A LM358
D4 1N4148
R29 10k
47k R30
4.7u C27
R4 100k
47n C28
R29 controls the 'decay' of the Noise-Gate by varying the cutoff frequency of the RC network in the envelope detector.
When R29 is at position (1), f_c = 0.31Hz
When R29 is at position (3), f_c = 3.39Hz
R30 and C28 form an additional Low-Pass Filter with a cutoff frequency of 72Hz.
INVERTING COMPARATOR WITH HYSTERESIS
R31 8.2k
RV5 100k
R32 10k
R29 controls the threshold (voltage) of the Noise-Gate.
When RV5 is turned to position (1): --> V_{ref} = 4.5V
When RV5 is turned to position (3): --> V_{ref} = 4.97V
4.7M R33
U2B LM358
When V_{in} > V_{ref}: V_{out} = 0V (LOW)
When V_{in} < V_{ref}: V_{out} = 9V (HIGH)
INDICATOR/DEBUGGING LEDS
2.2k R45
D11 LED
Q4 2N3904
22k R44
R48 100k
D7 1N4148
R47 2.2k
D12 LED
Q5 2N3904
22k R46
R49 100k
D8 1N4148
When the output of the hysteresis-comparator is 'HIGH' (approximately 9V), D11 lights up and D12 turns off.
When the output of the comparator is 'LOW' (approximately 0V), D11 turns off and D12 lights up.
The user can use those LEDs to get an idea of how 'sensitive' the comparator is and adjust its reference voltage through the RV5 potentiometer.
2.2k R40
4.7R Q3
D6 1N4148
R42 100k
Q3 2N3904
When the output of the hysteresis-comparator is 'HIGH', Q3 pulls the output line to GND.
Otherwise the output line is kept at high-impedance, because Q3 is in cutoff.
4.7k R34
R36 100R
R35 47k
D5 1N4148
Q2 2N3904
When the output of the hysteresis-comparator is 'HIGH', Q2 pulls the output line to 4.5V.
Otherwise the output line is kept at high-impedance, because Q2 is in cutoff.
NOISE GATE
OUTPUT STAGE
R37 470k
U5A LM358
R38 1k
C31 10u
R39 100k
CIRCUIT_OUT
4.7n C30
470n C23
1k R24
When Q2 and Q3 are in the cutoff region, the output buffer (U5B) is driven by the distortion-clipping pedal.
When Q2 and Q3 are saturated, the output stage is pulled 'LOW' and the guitar pedal is muted.
FIRST STAGE AMPLIFIER
R7 1k
R8 470k
U3A LM358
C9 47n
C11 470n
R9 100k
C13 250p
R11 470k
R12 10k
C14 68n
Q1 2N3904
R13 22R
R14 100k
C12 47n
R10 100k
This stage consists of a voltage buffer (U3A) followed by a common emitter amplifier with feedback (Q1).
The gain of this stage is approximately 19dB at 100Hz and rises gradually up to a maximum of 30dB for all audible frequencies above 1kHz.
DISTORTION AND CLIPPING
SECOND STAGE AMPLIFIER AND CLIPPING
This stage consists of a band-pass amplifier with adjustable gain and two clipping diodes.
R15 47k
U3B LM358
D1 1N4148
D2 1N4148
D3
RV3 100k controls the gain of the amplifier
When RV3 is at position (3), the op-amp acts as a simple voltage buffer (0 dB gain)
When RV3 is at position (1), the op-amp provides an extra 27dB gain for all frequencies between 100Hz and 10kHz
R16 100k
U4A LM358
100p C16
RV1 100k
R18 2.2k
C18 470n
C19 4.7n
R17 4.7k
470n C17
R16, C16 and C19 form a low-pass filter which attenuates frequencies above 10kHz more aggressively
TONE AND VOLUME CONTROL
R20 and C20 form a Low-Pass-Filter (LPF)
C20, R19 and R21 form a High-Pass-Filter (HPF)
20k R20
47n C21
RV2 22k
C20 15n
R19 2.2k
R21 10k
RV3 100k
R22 10k
C22 47n
R23 470k
U4B LM358
Potentiometer RV2 provides tone control by mixing the LPF and HPF signal paths
When RV2 is at position (1), low frequencies (bass) are favored over high frequencies
When RV2 is at position (3), high frequencies (treble) are favored over low frequencies
Potentiometer RV3 provides volume control.
When RV3 is at position (1), we get the maximum volume because the signal is fed directly to the output buffer U6A
When RV3 is at position (3), the output is muted because the output buffer U6A receives a constant 4.5V input.
POWER-SUPPLY:
DECOUPLING CAPACITORS AND
4.5V RAIL
D9: Reverse-polarity-protection Schottky diode for 9V battery input.
C1: Decoupling capacitor for 9V rail
D10 – R43: Power indicator LED with current limiting resistor
R1 – R2: Resistor network for generating the reference voltage of the 4.5V rail
U5A: Voltage buffer which acts as a low-impedance 4.5V voltage source.
C4 – C5: Decoupling capacitors for the 4.5V rail
+BATT
D9 1N5819
J1
C1 47u
R43 2.2k
D10 LED
R1 10k
R2 10k
U5B LM358
C2 47u
C3 100n
R3 10R
C4 47u
C5 100n
U1C LM358
C6 100n
U2C LM358
C10 100n
U3C LM358
C15 100n
U4C LM358
C25 100n
U5C LM358
C29 100n
M1 MountingHole_Pad
M2 MountingHole_Pad
M3 MountingHole_Pad
M4 MountingHole_Pad
Sheet: /
File: guitar_pedal.kicad_sch
Title:
Size: User
Date:
Rev:
KiCad E.D.A. 10.0.5
Id: 1/1